

# Bayesian Regression Analysis of Asset Returns Using Economic, Technical Indicators, and Competitor Asset Returns

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ISyE 6420 – Project

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1 December 2024

## Abstract

The accurate modeling and prediction of asset returns is a critical challenge in financial analysis, particularly under conditions of uncertainty and variability in competitor asset returns, economic indicators, and technical indicators (Gelman et al., 2004; Murphy, 2012). This project employs Bayesian regression to analyze the relationship between stock returns for JPMorgan Chase, Bank of America, and Citigroup, with a range of predictors, including macroeconomic variables (e.g., CPI, GDP) and technical indicators (e.g., RSI, moving average deviations). Using a dataset spanning from 1993 to 2023 sourced from the Bloomberg Terminal, the analysis integrates prior knowledge with observed data, enabling robust parameter estimation and uncertainty quantification.

The findings reveal significant correlations between key indicators and asset returns, with Bayesian methods providing credible intervals for model parameters and predictions. These insights emphasize the importance of Bayesian approaches in financial modeling, offering a framework to incorporate domain knowledge, manage uncertainty, and improve decision-making processes. This project demonstrates the versatility and efficacy of Bayesian analysis in addressing complex financial problems.

## 1 Introduction

### Context and Motivation

Modeling and predicting asset returns is a fundamental challenge in financial analysis, given the inherent complexity and uncertainty in financial markets. Accurate predictions are essential for decision-making in areas such as portfolio management, risk assessment, and trading strategies. However, traditional regression techniques often fail to effectively account for uncertainty and variability in key predictors such as macroeconomic indicators, technical metrics, and competitor asset returns. This project addresses the problem of understanding and modeling the relationship between these predictors and stock returns, with a focus on JPMorgan's, Bank of America's, and Citigroup's performances, using a Bayesian framework (Gelman et al., 2004).

The significance of this problem lies in the ability to manage uncertainty while making informed predictions. Financial markets are influenced by a multitude of dynamic factors, and quantifying the influence of these predictors on asset returns is crucial for enhancing predictive accuracy and robustness in decision-making processes.

### Objective

The primary objective of this project is to investigate the impact of macroeconomic and technical indicators of the three stock returns using Bayesian regression. By incorporating domain knowledge through priors and analyzing posterior distributions, the project aims to quantify the uncertainty

in model parameters and predictions. The ultimate goal is to derive insights that can improve the understanding of financial market dynamics and enhance predictive modeling techniques.

### Why Bayesian?

Bayesian methods are particularly well-suited for addressing the challenges inherent in financial modeling. Unlike traditional frequentist approaches, Bayesian analysis allows the incorporation of prior knowledge into the model, making it more robust in cases with limited or noisy data (Murphy, 2012). Additionally, Bayesian regression provides credible intervals for parameter estimates, offering a more nuanced understanding of the relationships between predictors and the target variable.

Furthermore, Bayesian methods excel in quantifying uncertainty, a critical aspect of financial predictions where outcomes are inherently probabilistic. This feature enables better risk assessment and confidence in predictions, making Bayesian approaches an invaluable tool for financial analysis. In this project, the use of Bayesian methods not only enhances the interpretability of the results but also provides a comprehensive framework for understanding the variability and dependencies in stock returns (van den Assem & Schouten, 2011).

## 2 Data Description

### Dataset Source

The dataset utilized for this analysis was derived from the Bloomberg Terminal, a widely used computer system that provides financial professionals with access to real-time financial data, news, and analytics. It was compiled into a CSV file named `market.csv`. The data spans from 1993 to 2023 and includes information on macroeconomic indicators, technical indicators, and adjusted closing prices of selected financial assets. This dataset provides a comprehensive foundation for analyzing the relationships between various predictors and asset returns (Feng et al., 2020).

### Key Variables

**Target Variable:** The primary target variable for this analysis is the daily asset returns for JPMorgan Chase (denoted as `jpm_return`), Bank of America (denoted as `bac_return`), and Citigroup (denoted as `citi_return`). Asset returns were calculated as the percentage change in adjusted closing prices between consecutive days.

### Predictors:

- **Economic Indicators:** Macroeconomic factors such as Consumer Price Index (CPI), Gross Domestic Product (GDP), Federal Funds Rate (FEDL01), and Unemployment Rate (UE) were included to capture broader economic trends.
- **Technical Indicators:** Variables such as Relative Strength Index (RSI) and deviations from moving averages (e.g., `jpm_20d_ma_dev`) were incorporated to account for market momentum and price trends. Additionally, lagged variables were generated to capture temporal dependencies (Feng et al., 2020).
- **Competitor Asset Returns:** In addition to JPMorgan, Bank of America, and Citigroup, other competitor banks such as Morgan Stanley and the S&P500 Financial Sector ETF (SPF) were included to provide a benchmark for assessing relative performance. These predictors aimed to capture interdependencies and spillover effects between major financial institutions, which may influence the returns of the target asset.

### Data Preprocessing

Data preprocessing was a critical step to ensure the quality and consistency of the dataset. The following steps were undertaken:

- **Handling Missing Values:** Rows containing missing values were removed to maintain the integrity of the analysis.
- **Standardization:** All numerical variables were standardized to have zero mean and unit variance, ensuring that variables on different scales contributed equally to the regression model.
- **Log Transformations:** For variables with non-positive values, a shifted log transformation was applied to address skewness and improve the normality of distributions.
- **Outlier Removal:** Percentile-based thresholds (1st and 99th percentiles) were used to identify and remove outliers, further improving data reliability.

The resulting cleaned and transformed dataset serves as a robust basis for Bayesian regression analysis, allowing for meaningful exploration of relationships between predictors and the target variable (Gelman et al., 2004).

### 3 Methodology

#### Model Description

The primary analytical approach used in this project was Bayesian linear regression, implemented using the PyMC library. The model is structured as follows:

- **Priors:** Weakly informative normal priors were assigned to the regression coefficients ( $\beta_i$ ) and the intercept ( $\alpha$ ), centered at 0 with a standard deviation of 1. A HalfNormal prior with a standard deviation of 1 was used for the noise term ( $\sigma$ ) to ensure positivity (Murphy, 2012).
- **Likelihood Function:** The likelihood of the observed returns ( $y$ ) was modeled as a normal distribution with a mean equal to the linear combination of predictors ( $X\beta + \alpha$ ) and standard deviation  $\sigma$ .
- **Posterior Inference:** Posterior distributions of the parameters were inferred using Markov Chain Monte Carlo (MCMC) sampling, specifically the No-U-Turn Sampler (NUTS), a gradient-based algorithm well-suited for high-dimensional parameter spaces (Gelman et al., 2004).

This Bayesian framework provides a comprehensive understanding of the relationships between predictors and the target variable while quantifying uncertainty through credible intervals for all estimated parameters.

#### Feature Engineering

Predictors were selected and engineered through a systematic process:

- **Selection:** An initial set of predictors was chosen based on domain knowledge, including macroeconomic indicators (CPI, GDP, UE, and FEDL01) and technical indicators (e.g., RSI, moving average deviations). Correlation analysis was conducted to identify the most influential variables for inclusion in the model (Feng et al., 2020).
- **Transformations:** Continuous predictors were standardized to ensure comparability and improve model convergence. Log transformations were applied to variables with skewed distributions to enhance normality. Lagged variables were created for economic indicators to account for delayed effects on the target variable, and deviations from moving averages were included to capture market trends.

These feature engineering steps ensured that the predictors were meaningful, interpretable, and effectively represented the underlying data-generating process (van den Assem & Schouten, 2011).

### Model Variants

To evaluate the contributions of different types of predictors, multiple model variants were tested:

- **Economic Indicators Only:** This model used only macroeconomic variables (CPI, GDP, UE, and FEDL01) to explain variations in asset returns, highlighting the broader economic influences.
- **Technical Indicators Only:** This variant relied exclusively on technical indicators (e.g., RSI, moving average deviations for each respective target variable) to capture market-specific dynamics (Feng et al., 2020).
- **Competitor Asset Returns:** Competitor banks to the respective target variable were included to provide a benchmark for assessing relative performance. These predictors aimed to capture interdependencies and spillover effects between major financial institutions, which may influence the returns of the target asset.
- **Mixed Model:** The final model incorporated competitor asset returns, economic indicators, and technical indicators to provide a comprehensive understanding of asset returns, leveraging the strengths of all predictor types (Murphy, 2012).

These variations allowed for comparative analysis, shedding light on the unique contributions of each predictor group and their combined effects on the target variable.

## 4 Results

### Posterior Summaries

The Bayesian models provided detailed insights into the relationships between predictors and asset returns. Posterior distributions for each coefficient were estimated, along with credible intervals, offering a robust understanding of parameter uncertainty. Key findings include:

- **Economic Indicators Only:** The posterior distributions for macroeconomic predictors such as CPI, GDP, and FEDL01 indicated significant contributions to the target variable, with tight credible intervals reflecting precise estimates (Gelman et al., 2004).
- **Technical Indicators Only:** Coefficients for variables such as RSI and moving average deviations (20d\_ma\_dev) demonstrated notable impacts on asset returns, albeit with slightly wider credible intervals compared to economic indicators.
- **Mixed Models:** The combined model leveraged both economic and technical indicators but exhibited some multicollinearity, which slightly widened the credible intervals for certain predictors.

Posterior distribution plots illustrate the spread and central tendencies of the estimates, confirming the significance of key predictors.

### Model Evaluation

**Performance Metrics:** The models were evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) for both train and test datasets. Additionally, posterior predictive checks, including scatterplots of observed vs. predicted values, were used to validate model performance.

- **JPM Analysis:**

- **Economic Indicators Only:** Train RMSE = 1.0483, Test RMSE = 1.0065, Train MAE = 0.7890, Test MAE = 0.7756.
- **Technical Indicators Only:** Train RMSE = 1.1060, Test RMSE = 1.0649, Train MAE = 0.8466, Test MAE = 0.8182.
- **Competitor Asset Returns Only:** Train RMSE = 1.3171, Test RMSE = 1.3027, Train MAE = 1.0326, Test MAE = 1.0293.
- **Mixed Variables:** Train RMSE = 1.3273, Test RMSE = 1.3122, Train MAE = 1.0398, Test MAE = 1.0387.

- **BAC Analysis:**

- **Economic Indicators Only:** Train RMSE = 1.0249, Test RMSE = 0.9607, Train MAE = 0.7690, Test MAE = 0.7389.
- **Technical Indicators Only:** Train RMSE = 1.0819, Test RMSE = 1.0106, Train MAE = 0.8247, Test MAE = 0.7849.
- **Competitor Asset Returns Only:** Train RMSE = 1.3027, Test RMSE = 1.2412, Train MAE = 1.0226, Test MAE = 0.9538.
- **Mixed Variables:** Train RMSE = 1.3099, Test RMSE = 1.2504, Train MAE = 1.0283, Test MAE = 0.9628.

- **Citi Analysis:**

- **Economic Indicators Only:** Train RMSE = 1.0260, Test RMSE = 0.9848, Train MAE = 0.7844, Test MAE = 0.7699.
- **Technical Indicators Only:** Train RMSE = 1.0691, Test RMSE = 1.0386, Train MAE = 0.8221, Test MAE = 0.8032.
- **Competitor Asset Returns Only:** Train RMSE = 1.3085, Test RMSE = 1.2389, Train MAE = 1.0277, Test MAE = 0.9787.
- **Mixed Variables:** Train RMSE = 1.3120, Test RMSE = 1.2447, Train MAE = 1.0318, Test MAE = 0.9781.

**Posterior Predictive Checks:** Scatterplots of observed vs. predicted values for train and test datasets revealed a strong alignment for models incorporating economic indicators. Models relying solely on technical indicators or competitor returns showed weaker predictive power, with greater dispersion around the 45-degree line.

## Key Insights

The analysis provided several significant insights:

- **Economic Indicators:** These predictors consistently outperformed others across all assets, reflecting the importance of macroeconomic trends in financial forecasting ([van den Assem & Schouten, 2011](#)).
- **Technical Indicators:** While impactful, these predictors were less consistent, possibly due to their short-term nature ([Feng et al., 2020](#)).
- **Competitor Returns:** Including competitor asset returns resulted in higher errors, suggesting limited predictive power in this context.
- **Mixed Models:** Although combining all predictor types offered a broader perspective, it introduced multicollinearity, which may have hindered overall performance.

## Discussion

### Analysis of Results

The analysis revealed meaningful relationships between the predictors and the target variables across different model variants. For models using economic indicators, variables like **CPI** and **FEDL01** demonstrated significant influence on asset returns, underscoring the impact of macroeconomic factors on market performance (Gelman et al., 2004).

Unexpected findings included the weaker-than-expected performance of the mixed models, which may be attributed to redundancy among predictors or limitations in capturing nonlinear interactions.

### Uncertainty Analysis

A key advantage of Bayesian methods was the ability to quantify uncertainty through credible intervals. Posterior distributions for regression coefficients provided valuable insights into the confidence of each predictor's effect. For instance, economic indicators displayed narrower credible intervals, indicating higher certainty in their influence, while technical indicators exhibited wider intervals, reflecting greater uncertainty.

### Model Limitations

Despite their strengths, the models have several limitations. The assumption of linear relationships between predictors and the target variable may oversimplify complex market dynamics. Additionally, potential overfitting in models with a large number of predictors, particularly in the mixed model, could limit generalizability. The relatively small dataset size may have constrained the models' ability to capture more nuanced patterns.

## Conclusion

### Summary

This study aimed to model and predict asset returns using Bayesian regression with various predictor types, including economic indicators, technical indicators, and competitor asset returns. By leveraging Bayesian methods, the analysis incorporated prior knowledge and provided credible intervals to quantify uncertainty (van den Assem & Schouten, 2011). Key findings demonstrated the significant influence of economic and technical indicators on asset returns, with economic indicators showing higher predictive stability.

### Future Work

Future work could extend this analysis by exploring hierarchical Bayesian models to capture group-level effects or incorporating more complex priors to encode domain-specific knowledge (Gelman et al., 2004). Nonlinear Bayesian models, such as Gaussian Processes, could provide a more flexible framework to capture interactions between predictors.

## References

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